

Week 1: TDD using JUnit5 and Mockito

Exercise 1: Setting Up JUnit

The screenshot shows an IDE with three main panels: Explorer, pom.xml, and Terminal.

Explorer Panel: Displays the project structure for 'COGNIZANT HANDSON'. The 'src' directory contains 'main' and 'test' subdirectories. The 'test' directory contains 'AppTest.java'. The 'pom.xml' file is selected in the Explorer.

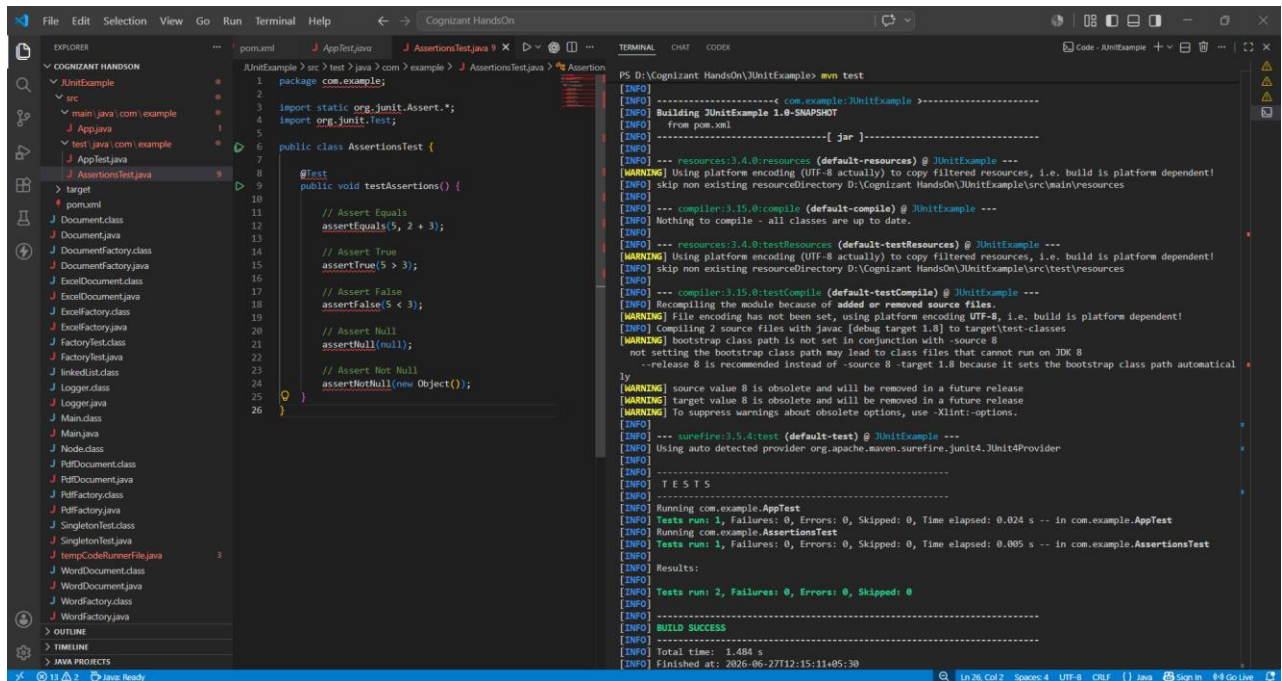
pom.xml Panel: Shows the contents of the 'pom.xml' file. The project is configured with the following details:

- groupId: com.example
- artifactId: JUnitExample
- version: 1.0-SNAPSHOT
- name: JUnitExample
- url: http://maven.apache.org
- dependencies: junit:junit:4.13.2 (scope: test)

Terminal Panel: Shows the output of the 'mvn test' command. The output indicates that the project is built successfully and the tests pass.

```
[INFO] -----[ jar ]-----
[INFO] --- resources:3.4.0:resources (default-resources) @ JUnitExample ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build
is platform dependent!
[INFO] skip non existing resourceDirectory D:\Cognizant HandsOn\JUnitExample\src\main\reso
urces
[INFO] --- compiler:3.15.0:compile (default-compile) @ JUnitExample ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- resources:3.4.0:testResources (default-testResources) @ JUnitExample ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build
is platform dependent!
[INFO] skip non existing resourceDirectory D:\Cognizant HandsOn\JUnitExample\src\test\reso
urces
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ JUnitExample ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- surefire:3.5.4:test (default-test) @ JUnitExample ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.025 s -- in com.e
xample.AppTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
[INFO] Total time: 1.611 s
[INFO] Finished at: 2026-06-27T12:04:20+05:30
[INFO] -----
PS D:\Cognizant HandsOn\JUnitExample>
```

Exercise 3: Assertions in JUnit



The screenshot shows an IDE with a project named 'Cognizant HandsOn'. The 'pom.xml' file is open, showing the following content:

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<groupId>com.example</groupId>
<artifactId>JUnitExample</artifactId>
<version>1.0-SNAPSHOT</version>
<packaging>jar</packaging>
<dependencies>
<dependency>
<groupId>org.junit.jupiter</groupId>
<artifactId>junit-jupiter</artifactId>
<scope>test</scope>
</dependency>
</dependencies>
</project>
```

The 'src/main/java/com/example' directory contains the following files:

- App.java
- AppTest.java
- AssertionsTest.java
- Document.java
- DocumentFactory.java
- ExcelDocument.java
- ExcelFactory.java
- FactoryTest.java
- LinkedList.java
- Logger.java
- Main.java
- Node.java
- PDFDocument.java
- PDFFactory.java
- SingletonTest.java
- TempCodeRunnerFile.java
- WordDocument.java
- WordFactory.java

The 'src/test/java/com/example' directory contains the following files:

- AppTest.java
- AssertionsTest.java
- Document.java
- DocumentFactory.java
- ExcelDocument.java
- ExcelFactory.java
- FactoryTest.java
- LinkedList.java
- Logger.java
- Main.java
- Node.java
- PDFDocument.java
- PDFFactory.java
- SingletonTest.java
- TempCodeRunnerFile.java
- WordDocument.java
- WordFactory.java

The 'AppTest.java' file contains the following code:

```
package com.example;

import static org.junit.Assert.*;
import org.junit.Test;

public class AppTest {

    @Test
    public void testApp() {
        // Assert Equals
        assertEquals(5, 2 + 3);

        // Assert True
        assertTrue(5 > 3);

        // Assert False
        assertFalse(5 < 3);

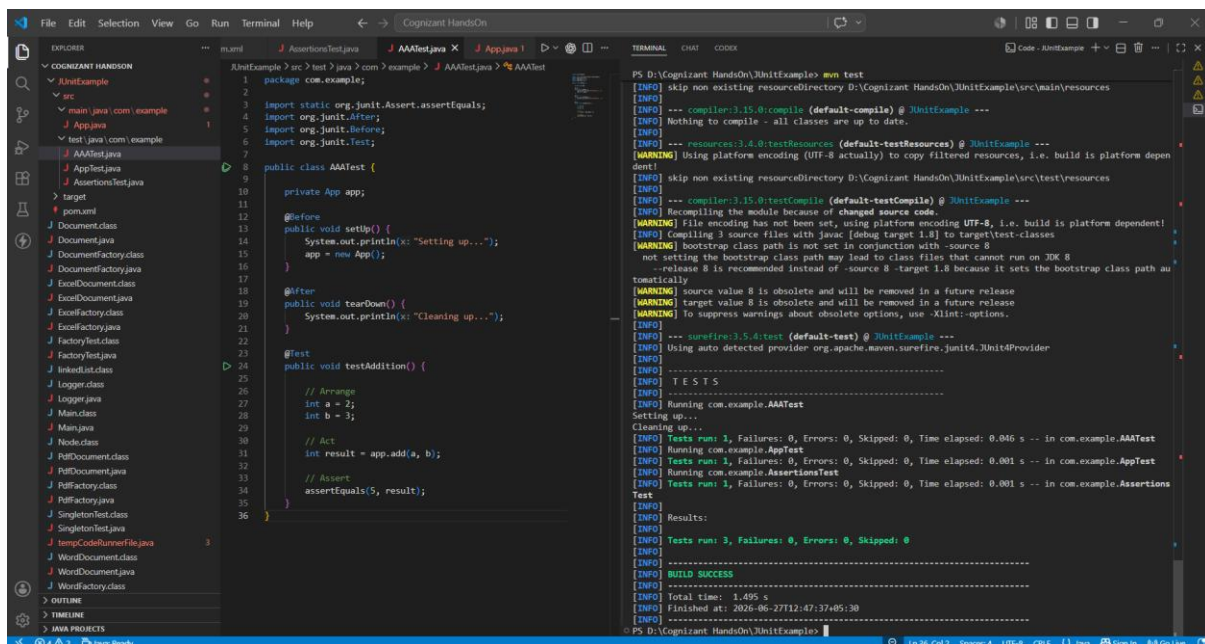
        // Assert Null
        assertNull(null);

        // Assert Not Null
        assertNotNull(new Object());
    }
}
```

The 'Terminal' window shows the following output:

```
PS D:\Cognizant HandsOn\JUnitExample> mvn test
[INFO] Scanning for projects...
[INFO] Building JUnitExample 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ JUnitExample ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\Cognizant HandsOn\JUnitExample\src/main/resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ JUnitExample ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ JUnitExample ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\Cognizant HandsOn\JUnitExample\src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ JUnitExample ---
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 2 source files with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
--release 8 is recommended instead of --source 8 -target 1.8 because it sets the bootstrap class path automaticall
ly
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ JUnitExample ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.024 s -- in com.example.AppTest
[INFO] Running com.example.AssertionsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.005 s -- in com.example.AssertionsTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.484 s
[INFO] Finished at: 2026-06-27T12:15:11+05:30
```

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods



The screenshot shows an IDE with a project named 'Cognizant HandsOn'. The 'pom.xml' file is open, showing the following content:

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<groupId>com.example</groupId>
<artifactId>JUnitExample</artifactId>
<version>1.0-SNAPSHOT</version>
<packaging>jar</packaging>
<dependencies>
<dependency>
<groupId>org.junit.jupiter</groupId>
<artifactId>junit-jupiter</artifactId>
<scope>test</scope>
</dependency>
</dependencies>
</project>
```

The 'src/main/java/com/example' directory contains the following files:

- App.java
- AppTest.java
- AssertionsTest.java
- Document.java
- DocumentFactory.java
- ExcelDocument.java
- ExcelFactory.java
- FactoryTest.java
- LinkedList.java
- Logger.java
- Main.java
- Node.java
- PDFDocument.java
- PDFFactory.java
- SingletonTest.java
- TempCodeRunnerFile.java
- WordDocument.java
- WordFactory.java

The 'src/test/java/com/example' directory contains the following files:

- AppTest.java
- AssertionsTest.java
- Document.java
- DocumentFactory.java
- ExcelDocument.java
- ExcelFactory.java
- FactoryTest.java
- LinkedList.java
- Logger.java
- Main.java
- Node.java
- PDFDocument.java
- PDFFactory.java
- SingletonTest.java
- TempCodeRunnerFile.java
- WordDocument.java
- WordFactory.java

The 'AppTest.java' file contains the following code:

```
package com.example;

import static org.junit.Assert.*;
import org.junit.Test;

public class AppTest {

    private App app;

    @Before
    public void setUp() {
        System.out.println("Setting up...");
        app = new App();
    }

    @After
    public void tearDown() {
        System.out.println("Cleaning up...");
    }

    @Test
    public void testAddition() {
        // Arrange
        int a = 2;
        int b = 3;

        // Act
        int result = app.add(a, b);

        // Assert
        assertEquals(5, result);
    }
}
```

The 'Terminal' window shows the following output:

```
PS D:\Cognizant HandsOn\JUnitExample> mvn test
[INFO] Scanning for projects...
[INFO] Building JUnitExample 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ JUnitExample ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform depen
dent!
[INFO] skip non existing resourceDirectory D:\Cognizant HandsOn\JUnitExample\src/main/resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ JUnitExample ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ JUnitExample ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform depen
dent!
[INFO] skip non existing resourceDirectory D:\Cognizant HandsOn\JUnitExample\src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ JUnitExample ---
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 3 source files with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
--release 8 is recommended instead of --source 8 -target 1.8 because it sets the bootstrap class path au
tomatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ JUnitExample ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.example.AppTest
[INFO] Setting up...
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.046 s -- in com.example.AppTest
[INFO] Running com.example.AssertionsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in com.example.AssertionsTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.495 s
[INFO] Finished at: 2026-06-27T12:47:37+05:30
```